Model-Based and Physics-Informed Deep Learning in Medical Imaging

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***Abstract*—Deep learning plays an increasingly important role in medical imaging, but purely data-driven models often lack interpretability and can behave unpredictably under distribution shift, which limits their clinical integration. Model-based deep learning (MB-DL) addresses these challenges by integrating physical models, algorithm-unrolling principles and optimization-based structure with neural networks, enabling more transparent, stable and data-efficient solutions. In this review, we discuss how these model-aware methods can enhance clinical relevance by surveying recent advances in MB-DL, including algorithm unrolling, physics-informed neural networks and hybrid mechanistic–neural architectures, and compare them with conventional black-box approaches across magnetic resonance imaging (MRI), computed tomography (CT), ultrasound and other modalities. We highlight how these methods improve explainability and outline the main open problems related to evaluation, robustness, uncertainty quantification and regulatory pathways. The aim is to provide a clear and structured overview of MB-DL and to clarify its emerging role in developing trustworthy medical-imaging AI systems.**

***Index Terms*—Model-based deep learning, algorithm unrolling, physics-informed neural networks, explainability, medical imaging.**

I. Introduction

Artificial Intelligence has become integral to medical imaging and diagnostic workflows, with recent advancements in deep neural networks (DNNs) significantly enhancing medical image processing capabilities. Across application domains such as image reconstruction, segmentation, and diagnostics, these models have achieved substantial success, often surpassing traditional analytical methods in both accuracy and automation. However, in safety-critical domains such as medical imaging, algorithmic outputs directly inform clinical decisions with potentially irreversible consequences. The opaque black-box nature of many purely data-driven models remains a major barrier to clinical trust, adoption, and regulatory approval. Accordingly, E. H. Houssein et al. [1] state that explainability is not a desirable extra but a fundamental requirement in this context. Classic analytical methods however lack robustness and perfromance. This dual failure (deep learn- ing lacks transparency while purely analytical methods lack robustness) has created an urgent clinical need for hybrid approaches.

To address this, an innovative paradigm called model-based deep learning (MB-DL), introduced by N. Shlezinger et al. [2], has emerged. This hybrid framework combines the rigorous domain knowledge of traditional physics-based models with the expressive power of data-driven learning. By integrating known forward physics and optimization-based structures directly into the network architecture, MB-DL requires significantly smaller training datasets and performs robustly even under imprecise knowledge of underlying physical models. This structural integration also yields models whose internal representations are directly interpretable in terms of known physical or algorithmic components.

Another closely related and prominent emerging approach is Physics-Informed Neural Networks (PINNs). Unlike standard DNNs that learn mappings purely from data, PINNs explicitly embed the governing physical laws that are typically formulated as partial differential equations (PDEs) directly into the neural network's loss function [3]. This enables the network to approximate solutions to complex biomedical problems, such as blood flow hemodynamics or magnetization dynamics in MRI, while ensuring the results remain physically consistent even when training data is sparse or noisy. By utilizing automatic differentiation, PINNs can solve both forward problems (simulating system behaviour) and inverse problems (uncovering hidden physical parameters), providing a mesh-free and flexible alternative to traditional numerical solvers.

This review highlights how methodologies based on MB-DL and PINNs offer a principled pathway to clinical trust by relying on intrinsic, structural interpretability rather than post-hoc explanations. Many established explainable AI (XAI) techniques, such as saliency maps and feature attribution methods, operate in a post-hoc manner, providing explanations only after a black-box model has generated its output. As noted by Huff et al. [4] these visualization techniques are widely used to interpret deep learning models in medical imaging. However, they often remain supplementary and can be unstable, in many cases failing to reveal the model's underlying mathematical logic.

In contrast, the interpretability of MB-DL arises from model transparency as defined by A. Q. Wang et al. [5]. These models are interpretable by design. In PINNs, transparency arises from the explicit enforcement of known physical laws, whereas in algorithm unrolling it stems from the correspondence between network layers and auditable steps of an iterative optimization algorithm. These architectures are designed such that every parameter and intermediate representation corresponds to a physically or mathematically meaningful quantity. By ensuring the model is understandable by construction, MB-DL improves transparency, controllability, auditability and consistency in clinically relevant settings.

This review article examines recent progress in explainable model-based deep learning, focusing on algorithm unrolling and physics-informed neural networks as principled alternatives to purely data- driven approaches. Sections II and III present their mathematical foundations and representative applications in MRI, CT, and ultrasound. Building on this, Section IV examines implications for explainability and clinical relevance, while Section V contrasts these approaches with conventional black-box methods. Finally, Section VI discusses the principal challenges that continue to impede clinical translation, including dataset bias, robustness, and regulatory considerations.

II. Model-Based Deep Learning and Algorithm Unrolling

This section outlines the principles of model-based deep learning (MB-DL) and its motivation in the medical imaging context. It then presents medical image reconstruction as a canonical inverse problem that is typically solved using iterative optimization. Finally, it focuses on algorithm unrolling, one of the most successful and widely adopted strategies in model-based deep learning, introducing its core theory and discussing its applications to MRI and CT reconstruction.

A. Model-Based Deep Learning

Model-based deep learning provides a physically and mathematically grounded framework for integrating the mathematical structure of classical inverse problems with the expressive power of modern neural networks. Rather than treating imaging as a generic pattern-recognition task, this paradigm embeds the known forward physics, acquisition operators, and statistical priors directly within the architecture or inference procedure. The result is a family of hybrid methods that combine interpretability and data efficiency with state-of-the-art performance. Figure 1 illustrates the continuous spectrum from purely data-driven to fully model-based approaches, with model-based deep learning positioned in the middle. This continuum demonstrates that greater incorporation of domain knowledge decreases the need for extensive training data, while adding learned components improves robustness to practical imperfections.

Two complementary methodologies have emerged as the dominant design patterns. The first category, model aided networks, constructs specialized deep neural network architectures by explicitly translating a classical iterative model-based algorithm, one that would be optimally employed given complete and accurate domain knowledge, into a finite depth layer-wise structure. Each iteration of the original algorithm is mapped to a corresponding neural network layer, while hand designed hyperparameters are promoted to trainable parameters. The resulting discriminative network preserves a clear interpretable correspondence between its layers and the analytical steps of the underlying model, yet becomes fully differentiable and amenable to end to end training. This strategy yields significant acceleration of inference speed, superior sample efficiency, and enhanced robustness to model mismatch. The second category, denoted DNN aided inference (sometimes referred to as neural augmentation), retains a traditional provably convergent model-based algorithm as the primary inference engine and selectively replaces or augments only those specific sub modules that are analytically intractable or highly sensitive to modeling errors with learned deep neural networks. The overall processing pipeline thus remains dominated by well understood domain knowledge driven operations that preserve theoretical guarantees and high interpretability, while the inserted DNN components capture complex statistical structures or compensate for incomplete physics and unknown nuisance parameters. For practical system design for medical imaging, the community has converged on the following widely adopted three step guideline, as expressed in subsequent foundational works on Shlezinger et al. [2]:

1. Verify the availability of exploitable domain knowledge (forward physics, measurement operators, anatomical or physiological priors) along with sufficient labeled or high fidelity simulation generated training data.
2. Identify and start from a suitable classical iterative model-based algorithm that would perform well under the idealized assumption of perfect model knowledge.
3. Incorporate deep learning precisely and only at the principal bottleneck of the chosen algorithm, typically its acute sensitivity to modeling errors, unknown parameters, or oversimplified assumptions.

By adhering to these three principles, model-based deep learning retains the interpretability, data efficiency, and theoretical foundation of the original model-based method while allowing the neural network components to learn the missing or uncertain parts from data, thereby achieving substantially greater robustness, reconstruction quality, and diagnostic performance than either purely model-based or purely data driven alternatives.

B. Inverse Problems in Clinical Imaging: The Natural Role of Iterative Optimization

Medical imaging reconstruction tasks are naturally formulated as inverse problems, in which the acquired measurement data y are related to the underlying image x through a forward model

$$y = A(x) + n, \tag{1}$$

where A represents the forward operator encoding the acquisition physics, and n denotes measurement noise and other modeling errors. Direct inversion of A is generally impossible or unstable, as the operator is often ill-conditioned, underdetermined, or both. High quality reconstructions are therefore obtained by solving a regularized optimization problem of the form

$$\hat{x} = \arg\min_{x} \frac{1}{2}\|y - A(x)\|_2^2 + \lambda R(x), \tag{2}$$

where the data-fidelity term enforces consistency with the measurements and $R(x)$ encodes prior information about the image. Such optimization problems are typically solved using iterative algorithms, including gradient-based methods,

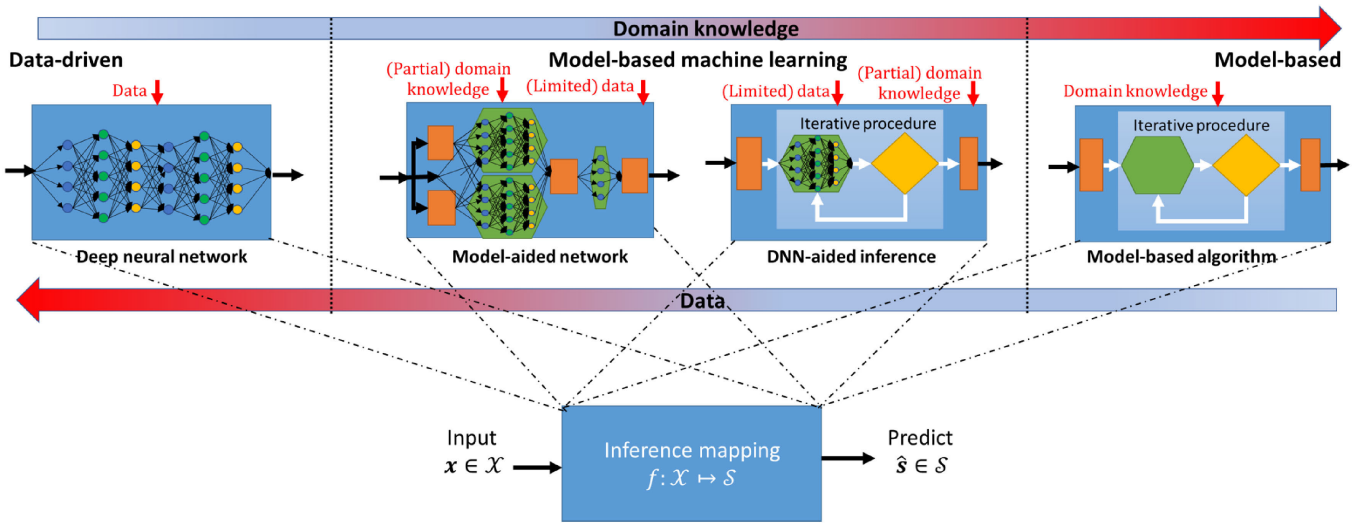


Fig. 1. Illustration of the continuous spectrum between model-based and data-driven inference, with fully model-based methods and deep learning at the opposite extremes. [2]

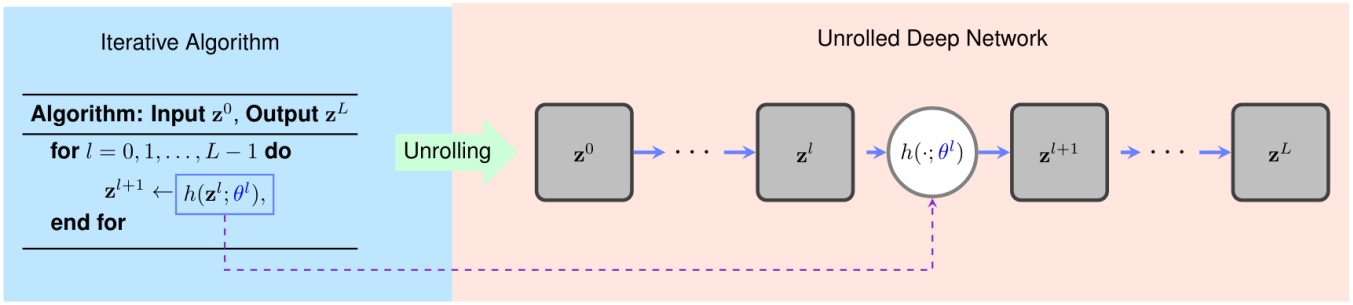


Fig. 2. Illustration of algorithm unrolling. Iterations are stacked to build neural network. [6]

proximal algorithms, primal-dual schemes, and the Alternating Direction Method of Multipliers (ADMM) [7].

Table I summarizes representative forward operators encountered in common clinical imaging modalities.

TABLE I
COMMON INVERSE PROBLEMS AND THEIR FORWARD OPERATORS IN CLINICAL IMAGING

Application	Forward operator A
Image denoising	$A = I$ (identity)
Super-resolution	$A = SB$ (S downsampling, B blur)
Parallel / accelerated MRI	$A = PFS$ (P coil sensitivities, F Fourier, S subsampling)
Sparse-view / low-dose CT	$A = R$ (system matrix / Radon)

These forward operators are derived from well-established physical models. Classical iterative solvers have therefore dominated clinical reconstruction pipelines for decades. They provide strong interpretability and do not require annotated training data. However, in practice, their performance is highly sensitive to inaccuracies in the forward operator A, such as imperfect coil sensitivity maps, patient motion, or beam-hardening effects, as well as to mis-specification of the regularization parameter λ and the functional $R(\cdot)$. Even minor mismatches can introduce severe artifacts that are readily recognized by clinicians as unreliable. This sensitivity, combined with growing clinical demands for faster acquisitions and reduced radiation dose, motivates reconstruction approaches that retain the physical consistency and interpretability of classical algorithms while enhancing robustness through data-driven learning. The following section illustrates how algorithm unrolling addresses these challenges.

C. Algorithm Unrolling

Algorithm unrolling addresses these limitations by mapping each iteration of a classical solver to a corresponding layer in a deep neural network, resulting in a trainable and fully differentiable architecture that preserves the structure and interpretability of the original algorithm.

A canonical example is the Iterative Shrinkage-Thresholding Algorithm (ISTA), widely used for sparse coding and introduced in its learned form by Gregor and LeCun [8]. Each ISTA iteration consists of a linear update followed by a nonlinear shrinkage operation, which naturally resembles a neural network layer. By unrolling a fixed number of ISTA iterations and promoting step sizes, thresholds and regularization parameters to trainable variables, one obtains

the Learned ISTA (LISTA) network. LISTA preserves a clear correspondence between network layers and algorithmic steps, while achieving substantially faster convergence and improved sample efficiency compared to its classical counterpart.

This unrolling principle generalizes to a broad class of iterative methods, including gradient descent, proximal algorithms, ADMM [7] and primal-dual schemes [9]. In these architectures, hand-crafted hyperparameters are replaced by learned parameters, allowing the network to adapt to imperfect forward models and compensate for modeling errors while maintaining explicit data-consistency and regularization steps. A schematic illustration of algorithm unrolling is shown in Figure 2. Algorithm unrolling repeatedly applies a fixed analytic update rule, denoted as the function h. By mapping each iteration of this rule to a single network layer and stacking a finite number of such layers, algorithm unrolling constructs a deep neural network. The parameters θ^l in each layer l (which typically correspond to regularization coefficients, step sizes, or other hyperparameters in the original iterative algorithm) are made trainable. This allows the network to adapt optimally to data while remaining human interpretable, as each parameter retains a clear connection to its role in the underlying optimization process.

Algorithm unrolling has been successfully applied in a range of imaging and signal processing tasks, including sparse coding networks for high-resolution image recovery [8] and sparse biological signal localization [10], where sources are blurred by the point-spread function of the imaging system. By embedding algorithmic structure directly into the network, each layer admits a clear interpretation in terms of data fidelity, regularization, or sparsity enforcement. This property makes unrolled networks particularly well suited for clinical imaging applications, where trust, interpretability and regulatory considerations are essential.

In summary, algorithm unrolling bridges the gap between classical model-based solvers and modern deep learning, combining interpretability, computational efficiency and robustness within a unified framework.

D. Algorithm Unrolling for Clinically Robust MRI and CT Reconstruction

Algorithm unrolling is particularly effective in medical imaging, where undersampled or low-dose acquisitions make classical reconstruction methods ill-posed and computationally demanding. By embedding iterative optimization steps into network layers, unrolled architectures combine the interpretability of model-based reconstruction with the adaptability of data-driven learning.

1) Interpretability and Parameter Correspondence in Unrolled Networks: A defining strength of algorithm unrolling is that it preserves a direct, auditable link between learned parameters and specific steps of the underlying optimization algorithm,a property largely absent in conventional end-to-end convolutional neural networks. Because each network layer represents one iteration of a classical solver, such as ISTA, ADMM, proximal gradient descent,or primal-dual hybrid gradient methods, every trainable parameter retains a clear physical and mathematical interpretation.

This correspondence is easiest to see in the proximal gradient descent (PGD) family of algorithms, which forms the basis of many unrolled MRI and CT reconstructions. The canonical PGD update,

$$x^{k+1} = \mathrm{prox}_{\gamma\lambda R}\big(x^k - \gamma\nabla f(x^k)\big), \tag{3}$$

combines an explicit gradient step on the data-fidelity term f (e.g., $\|y-Ax\|_2^2$) with a proximal evaluation of the regularizer R. When this iteration is unrolled, the step size γ_l and regularization weight λ_l in layer l become trainable scalars that directly control how aggressively the network enforces measurement consistency versus prior knowledge.

An equivalent form, popular in Learned ISTA (LISTA) and its descendants, rewrites the update as

$$x^{k+1} = S_{\lambda/\mu}\left(\left(I - \frac{1}{\mu}W^TW\right)x^k + \frac{1}{\mu}W^Ty\right), \tag{4}$$

where the parameter μ_l now plays the role of an adaptive step size (often initialized as the inverse Lipschitz constant of the forward operator). In both formulations, larger learned step sizes (γ_l or $1/\mu_l$) in early layers accelerate coarse removal of aliasing or streak artifacts, while smaller values in later layers protect fine anatomical details from over-regularization.

Learned transforms or proximal operators, often implemented using compact CNN modules or linear filters, replace hand-crafted denoisers while still explicitly representing the regularization step of the optimization. In augmented Lagrangian formulations, learned penalty parameters enforce consensus between data fidelity and prior terms, and their evolution across layers reveals how the reconstruction departs from the raw measurements.

Because these parameters preserve their original algorithmic meaning, clinicians and regulators can visualize intermediate reconstructions at each layer, diagnose failure modes, compare learned parameters with theoretically expected ranges, and, in some cases, adjust specific parameters for individual patients without retraining the entire network. This transparency improves clinical confidence, generalization across scanners, and regulatory viability.

2) MRI Reconstruction: In MRI, undersampled k-space measurements complicate inversion. Yang et al. [11] introduced ADMM-CSNet, which unrolls the ADMM algorithm and learns both sparsifying transforms and regularization parameters from data. Compared with traditional iterative methods, ADMM-CSNet achieves comparable reconstruction quality using as little as 10% of the measurements, while reducing reconstruction time by approximately a factor of 40. Hosseini et al. [12] further improved performance by unrolling proximal gradient descent with dense connections inspired by momentum-based acceleration.

3) CT Reconstruction: In CT, low-dose or sparse-view acquisitions increase sensitivity to noise and artifacts. Adler et al. [9] unrolled the primal-dual hybrid gradient algorithm, replacing proximal operators with CNN-based modules and jointly learning algorithmic and operator parameters. Wu et al. [13] extended this approach by integrating a detection network and jointly optimizing reconstruction and detection tasks. The layer-wise structure ensures each operation remains interpretable, allowing clinicians to trace contributions to the final image.

These examples demonstrate how algorithm unrolling combines computational efficiency, interpretability, and clinical relevance. By embedding physical models and optimization structure into trainable architectures, unrolled networks bridge classical model-based theory and practical clinical application.

III. Physics-Informed Neural Networks

Modern medical imaging relies fundamentally on physics. Whether reconstructing magnetic resonance images, modelling tissue deformation, or simulating contrast kinetics, the underlying processes are governed by partial differential equations (PDEs) [14]. Traditional numerical solvers, such as the Finite Element Method (FEM), struggle when data is noisy or sparse, while standard deep learning models often ignore the underlying physical laws entirely. Physics-Informed Neural Networks (PINNs) bridge this gap by seamlessly integrating data with governing physical equations [3], [15]. Instead of relying on discrete grids, PINNs approximate solutions using continuous functions, offering a mesh-free alternative well-suited for complex biomedical problems.

A. Mathematical Framework

The core innovation of the PINN framework lies in the construction of a loss function that strictly enforces physical consistency [15]. In its most general form, a nonlinear PDE given by

$$u_t + \mathcal{N}[u;\lambda] = 0, \quad x \in \Omega, \quad t \in [0,T], \tag{5}$$

where $u(t,x)$ is the unknown solution and $\mathcal{N}[\cdot;\lambda]$ is a nonlinear operator parameterized by λ. Raissi et al. [3] define a physics-informed residual network $f(t,x)$ regarding Eq. (5) as:

$$f := u_t + \mathcal{N}[u;\lambda]. \tag{6}$$

A deep neural network $\hat{u}(t,x;\theta)$ approximates the solution $u(t,x)$. The network parameters θ are learned by minimizing a composite loss function

$$\mathcal{L}_{total} = w_{data}\mathcal{L}_{data} + w_{PDE}\mathcal{L}_{PDE}, \tag{7}$$

where the individual components

$$\mathcal{L}_{data} = \frac{1}{N_u}\sum_{i=1}^{N_u}\left|\hat{u}(t_u^i, x_u^i;\theta) - u^i\right|^2, \tag{8}$$

$$\mathcal{L}_{PDE} = \frac{1}{N_f}\sum_{j=1}^{N_f}\left|f(t_f^j, x_f^j;\theta)\right|^2, \tag{9}$$

drive the network to fit the observed measurements and the physical laws, respectively. Here, $\mathcal{L}_{data}$ ensures the prediction matches available training data, while $\mathcal{L}_{PDE}$ penalizes deviations from the physics at randomly sampled collocation points. This dual-objective formulation forms the foundation of PINNs, ensuring learned solutions strictly obey physical laws [3].

B. Automatic Differentiation

To compute the residual in Eq. (6) effectively, the framework utilizes Automatic Differentiation (AD). Unlike numerical differentiation, which introduces truncation errors, AD computes derivatives exactly by applying the chain rule to the neural network graph. This property is essential as the PDE residual often depends on high-order derivatives, which are sensitive to noise and discretization. Using automatic differentiation ensures that the physics constraints are enforced with machine-precision accuracy rather than approximate finite-difference schemes. Consequently, AD leads to more stable and reliable training.

C. Forward and Inverse Problems

The versatility of PINNs allows them to address two distinct classes of challenges often encountered in medical physics.

1) Forward Problems: In a forward problem, the physical parameters are known, and the goal is to simulate the resulting system behaviour. For example, in MRI, a forward problem corresponds to predicting signal evolution given known tissue properties and pulse sequence parameters. PINNs solve this by finding the solution $u(t,x)$ that satisfies the PDE alongside given initial and boundary conditions [16].

2) Inverse Problems: Conversely, an inverse problem attempts to uncover the cause from the effect. In the MRI context, this implies recovering unknown tissue properties from limited and often noisy measurements collected during the scan [17]. PINNs excel here by simultaneously inferring the solution $u(t,x)$ and identifying the unknown parameters λ in Eq. (5). This capability allows for the data-driven discovery of governing laws and physical properties while keeping solutions consistent with the underlying MRI physics.

D. Case Study: Incompressible Navier-Stokes Equation

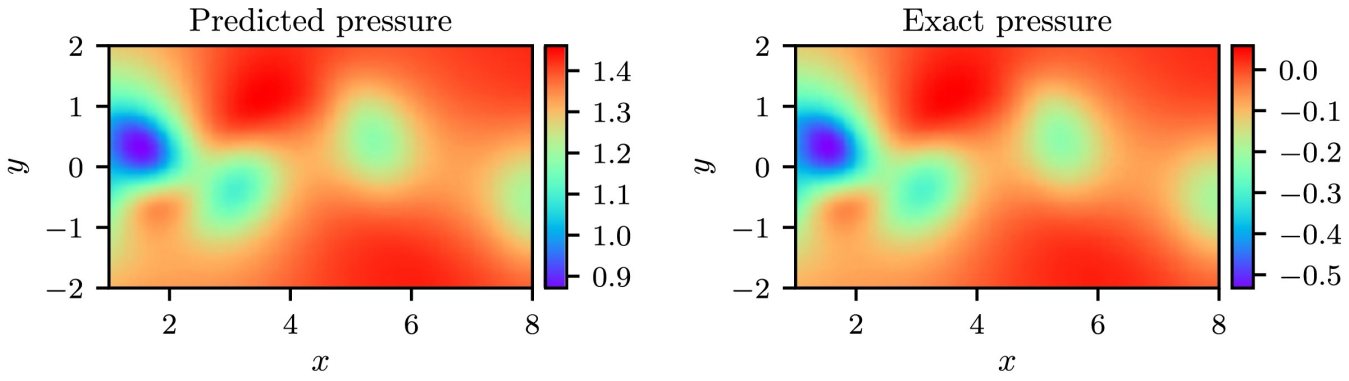
To illustrate the potential of PINNs for inverse problems, the reconstruction of fluid flow is examined, which is directly applicable to biomedical hemodynamics [15]. Raissi et al. [3] demonstrated this using the Navier-Stokes equations for incompressible flow. The governing equations for velocity (u,v) and pressure p are

$$u_t + (uu_x + vu_y) = -p_x + \frac{1}{Re}(u_{xx} + u_{yy}), \tag{10}$$

$$v_t + (uv_x + vv_y) = -p_y + \frac{1}{Re}(v_{xx} + v_{yy}), \tag{11}$$

$$u_x + v_y = 0. \tag{12}$$

where Re represents the Reynolds number. In this case study, the pressure p is treated as a latent variable with no available



Correct PDE	$u_t + (uu_x + vu_y) = -p_x + 0.01(u_{xx} + u_{yy})$ $v_t + (uv_x + vv_y) = -p_y + 0.01(v_{xx} + v_{yy})$
Identified PDE (clean data)	$u_t + 0.999(uu_x + vu_y) = -p_x + 0.01047(u_{xx} + u_{yy})$ $v_t + 0.999(uv_x + vv_y) = -p_y + 0.01047(v_{xx} + v_{yy})$
Identified PDE (1% noise)	$u_t + 0.998(uu_x + vu_y) = -p_x + 0.01057(u_{xx} + u_{yy})$ $v_t + 0.998(uv_x + vv_y) = -p_y + 0.01057(v_{xx} + v_{yy})$

Fig. 3. Comparison of exact and learned pressure fields [3].

training data. The neural network takes coordinates (t, x, y) as input and outputs (u, v, p), minimizing the residuals of Eqs. (10)–(12).

Figure 3 illustrates the predicted and exact pressure fields together with the corresponding velocity field patterns. The visualization demonstrates that despite having access only to sparse velocity observations, the network successfully reconstructs the pressure field with high fidelity. This showcases the potential of PINNs for physics-constrained inverse problems in biomedical imaging [14].

This result is particularly relevant for cardiovascular medicine, where direct measurement of intravascular pressure fields is invasive and clinically challenging, while blood velocity measurements are considerably more feasible in practice [3], [15].

IV. Explainability and Clinical Relevance in Medical Imaging

To fully evaluate the role of explainability in medical AI, it is first necessary to distinguish between its different forms and to clarify how these forms relate to model architecture. While the objective of Explainable Artificial Intelligence (XAI) is to render model behavior understandable to clinicians and regulatory bodies, the mechanisms by which this objective is achieved differ substantially between opaque black-box approaches and model-aware or model-based designs.

Building on this distinction, this section first reviews the main forms of explainability and their relevance in medical imaging. It then provides an overview of common medical imaging modalities and examines how artificial intelligence can be integrated to enhance their performance. The discussion subsequently narrows to magnetic resonance imaging (MRI) as a representative and clinically important modality. Finally, the section focuses on the application of model-based deep learning (MB-DL) and PINNs to MRI reconstruction and analysis, highlighting how model transparency supports clinical relevance and trust.

A. Explainability Trough Model Transparency

E. H. Houssein et al. [1] have successfully argued that interpretability is a prerequisite for clinical adoption. In the medical domain, interpretability is defined as the model's ability to provide a human-understandable explanation for its decisions, allowing clinicians to verify, trust, and act upon the results. To understand how we arrive at this state, a fundamental distinction must be made between post-hoc and intrinsic-structural interpretability, as proposed by D. T. Huff et al. [4].

Post-hoc explainability is typically paired with traditional black-box deep learning models and seeks to explain a model's reasoning only after a decision has been made. Common approaches include saliency maps, Grad-CAM, and feature attribution methods. E. H. Houssein et al. [1] show that although these techniques are widely used, they are often unstable: small perturbations in the input can lead to substantially different explanations, and it is frequently unclear whether the resulting visualization reflects the model's true decision logic or merely hallucinates a plausible justification for an incorrect output.

In contrast, the explainability addressed by Physics-Informed Neural Networks (PINNs), model-based neural networks, and algorithm unrolling is intrinsic rather than post-hoc. These approaches embed physical laws, forward models,

or iterative optimization procedures directly into the network architecture, yielding models whose components, parameters, and intermediate representations have clear mathematical or physical meaning.

In this setting, interpretability arises from the model design itself: network parameters often correspond to physically meaningful quantities, regularization terms, or algorithmic steps, and the inference process mirrors established reconstruction or inverse-problem formulations. Rather than explaining why a black-box model produced a specific output, these architectures aim to ensure that the model is understandable by construction, thereby improving transparency, controllability, and trustworthiness in clinically relevant settings.

B. Elements of Explainability in Medical Imaging

Because there are numerous paths to achieving explainability, A. Q. Wang et al. [5] proposed a general framework for identifying the necessary elements of explainability based on specific clinical tasks. This framework defines five core elements. Localizability refers to the ability to identify where in the image the model is focusing its attention, while visual recognizability describes the extent to which the features used by the model correspond to human-recognizable anatomy or pathology. Physical attribution concerns the linkage between model outputs and known physical properties, such as T1 or T2 relaxation times in MRI. Transparency captures the clarity of the model's internal computational logic, and actionability denotes the extent to which interpretable outputs translate directly into meaningful clinical insights and decision-making

Within this framework, the first three elements—localizability, visual recognizability, and physical attribution—are often achieved through post-hoc explainability methods. Transparency, however, is an intrinsic property of the model itself. By prioritizing intrinsic-structural interpretability, model-based deep learning inherently fulfills the requirement for transparency, thereby providing a more robust foundation for actionability than purely data-driven, black-box approaches.

C. Applications in Medical Imaging

There are numerous medical imaging modalities, each with its own advantages, limitations, and underlying physical principles. Among the most widely used are Magnetic Resonance Imaging (MRI), Computed Tomography (CT), and Ultrasound (US). CT employs X-ray attenuation to produce high-resolution cross-sectional images, particularly effective for visualizing dense structures such as bones but constrained by ionizing radiation exposure. Ultrasound, based on acoustic wave propagation and reflection, offers real-time imaging with excellent temporal resolution, though it is often affected by noise and limited spatial detail. MRI, by contrast, relies on nuclear magnetic resonance (NMR) phenomena, using magnetic fields and radiofrequency pulses to generate rich soft-tissue contrast and diverse quantitative information.

For the purposes of this review, MRI stands out as the most promising modality for advanced AI methodologies. Its signal formation is governed by well-understood physics, making it especially suitable for physics-informed approaches such as PINNs and for model-based reconstruction methods rooted in domain knowledge. At the same time, MRI reconstruction typically involves solving complex optimization problems, creating a natural setting for algorithm unrolling techniques. Furthermore, MRI remains one of the more time-consuming and costly imaging procedures; therefore, improvements in image quality, acquisition speed, or reconstruction efficiency translate directly into substantial clinical and economic benefits.

Building on the insights provided by M. Ahmad et al. [14] and E. H. Houssein et al. [1], we can now direct our attention to the main applications of XAI in MRI imaging, which can be summarized as follows:

- Image Reconstruction: MRI scanners naturally acquire data in the frequency domain (known as k-space), which must be transformed into an image. Deep learning models learn how to fill in missing or undersampled data, enabling faster scans while still producing sharp, clinically reliable images.
- Image Segmentation and Quantification: Once an image is reconstructed, CNNs and transformer-based models can automatically delineate organs, tissues, or lesions. This provides precise measurements of volumes, shapes, or tissue properties, greatly reducing the manual workload for radiologists and improving consistency.
- Diagnosis and Prognosis: Radiomics and deep learning methods can extract subtle patterns in MRI images that are often imperceptible to the human eye to detect diseases earlier, determine their severity, or predict future outcomes. These approaches support clinicians by offering data-driven insights for diagnosis and treatment planning.

Among these applications, image reconstruction has seen the most direct integration of physics-informed neural networks (PINNs) and algorithm unrolling frameworks. Image-Segmentation/quantification and Diagnosis/prognosis demand different forms of explainability, such as *localizability* and *visual recognizability*. These aspects are typically supported by techniques like saliency maps or attention-based visualizations, though a deeper exploration of these methods lies beyond the scope of this review.

D. MRI Image Reconstruction

MRI image reconstruction presents a central challenge in medical imaging because the scanner does not acquire images directly in the spatial domain. Instead, measurements are collected in k-space, the frequency-domain representation of the object, determined by the physics of signal encoding and receiver coil sensitivities. Image reconstruction involves inverting this process, typically via an inverse Fourier transform, to recover the spatial-domain image.

To reduce scan times and improve patient comfort, k-space measurements are typically undersampled. This means fewer measurements are collected than required by the Nyquist

criterion. The Nyquist criterion states that the sampling rate must be high enough to capture the highest spatial frequencies. When it is not met, undersampling produces aliasing artifacts. These are alternative image patterns consistent with the acquired data. The result is blurring and reduced image quality. Traditional reconstruction techniques, such as compressed sensing and parallel imaging, regularize this problem by imposing sparsity priors or using coil sensitivity maps. However, these methods often trade off between image sharpness and artifact suppression, and their performance is limited by handcrafted priors.

Deep learning-based reconstruction methods have demonstrated substantial improvements in image quality and reconstruction speed. However, purely data-driven models often suffer from limited interpretability and reduced generalization across scanners, pulse sequences, and anatomical variability [14]. These limitations motivate the development of hybrid approaches that combine physical modeling with data-driven learning, such as algorithm unrolling and physics-informed neural networks.

E. Physics-Informed and Model-Based Deep Learning for MRI Reconstruction

MRI reconstruction is particularly well suited to physics-informed and model-based deep learning approaches, as it combines well-understood signal formation physics with challenging inverse problems arising from undersampled k-space acquisitions. The techniques discussed in the previous sections naturally converge in this setting.

Model-based deep neural networks integrate classical MRI reconstruction frameworks with learned components, such as data-adaptive sparsity priors or learned coil sensitivity representations. By embedding these learned elements within established reconstruction pipelines, such approaches preserve interpretability while benefiting from the flexibility of data-driven learning. Shlezinger et al. [2] exemplify this paradigm by demonstrating how model-aware network design can achieve improved reconstruction quality with reduced training data requirements.

Algorithm unrolling provides a complementary strategy by explicitly mapping iterative MRI reconstruction algorithms onto neural network architectures. As shown by Monga, Li, and Eldar [6], each iteration of a classical solver used to recover images from undersampled k- space data can be represented as a network layer, with algorithmic parameters learned from data. This design preserves the structure of the original reconstruction algorithm while enabling end-to-end training. As a result, unrolled networks achieve faster convergence, reduced aliasing and undersampling artifacts, and improved reconstruction quality compared to traditional iterative methods.

Physics-informed neural networks (PINNs) take a more explicit approach to incorporating MRI physics by embedding the governing equations of signal formation directly into the training objective. By enforcing consistency with the dynamics of magnetization and k-space acquisition, PINNs ensure that reconstructed images respect both the measured data and the underlying physical laws. For example, Ahmed et al. [17] proposed a PINN architecture that incorporates the Bloch equations, which describe magnetization dynamics, into the loss function. This physics-constrained formulation improves robustness and interpretability while maintaining competitive reconstruction performance.

Taken together, model-based neural networks, algorithm unrolling, and physics-informed neural networks offer complementary mechanisms for incorporating physical knowledge into MRI reconstruction. By balancing data-driven learning with explicit physical and algorithmic structure, these approaches yield reconstructions that are more accurate, physically consistent, and robust to variations in acquisition protocols.

V. Comparison and Research Gaps

Deep learning has transformed medical imaging, yet its performance depends strongly on the structure and assumptions underlying the model. In current practice, two families dominate: purely data-driven networks and model-based deep learning (MB-DL) architectures. Both offer unique strengths, but they also introduce different types of risk. Understanding these differences is essential for evaluating reliability, robustness and clinical readiness. This section reviews the limitations of purely data-driven approaches and motivates why model-based and hybrid strategies have emerged as promising alternatives.

A. Limitations of Purely Data-Driven Models

Purely data-driven networks such as CNNs and vision transformers can achieve high performance when trained on large and representative datasets. Their main strength lies in flexibility: they learn complex mappings directly from data, without relying on explicit knowledge of imaging physics. However, this same flexibility introduces important weaknesses that limit their reliability in clinical practice. These weaknesses become most apparent when training data are limited or when models are deployed outside the conditions observed during training, a well-documented issue in medical imaging [18], [19].

A first challenge is limited generalization under domain shift. Medical datasets are often small, unevenly sampled, or biased toward specific scanners, institutions, or patient populations. Under such conditions, purely data-driven models tend to overfit and perform poorly when acquisition settings change. Clinically, this implies that a model trained in one hospital may behave unpredictably when deployed in another, undermining trust in its outputs. This phenomenon has been observed across multiple imaging modalities and diagnostic tasks [20]. In addition, interpretability degrades under distribution shift: common post-hoc explainability tools, such as saliency maps or attention visualizations, often change substantially even when predictions remain similar, making clinical validation and failure analysis difficult.

Figure 4 illustrates this sensitivity to dataset shift. Images acquired under different physical imaging conditions form distinct clusters in representation space, even when low-level

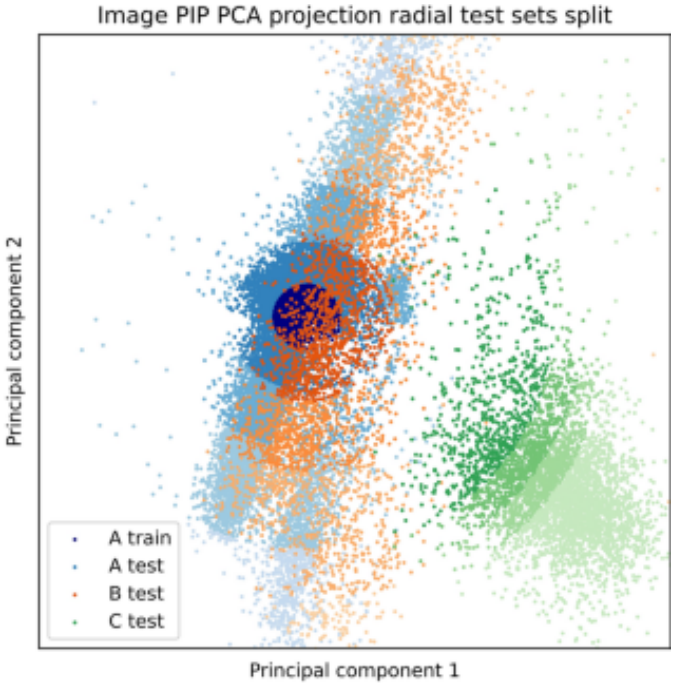


Fig. 4. Dataset shift effects in purely data-driven medical imaging models. Low-dimensional PCA embedding of image representations for training and external test datasets. Adapted from [21].

visual differences are subtle. This highlights that purely data-driven models rely heavily on correlations specific to the training distribution, rather than on invariant physical structure. As a result, performance can degrade sharply under realistic clinical variability.

A second limitation is the lack of physical constraints. Without explicit knowledge of anatomy or imaging physics, data-driven models may hallucinate structures or produce reconstructions that violate basic physical principles. Error analysis is therefore difficult, since internal representations are opaque and the model provides no indication of which regions of the output are unreliable. These issues are amplified when measurements are sparse or corrupted, as the network must infer both the structure of the problem and the final solution directly from data. Taken together, these limitations motivate the integration of physical models or algorithmic structure into learning-based systems. Model-based and hybrid approaches aim to constrain learning with domain knowledge, improve interpretability, and deliver more predictable behaviour under realistic clinical conditions.

B. Foundational Strengths of Model-Based Deep Learning

Model-based deep learning differs from purely data-driven approaches by embedding physical models or algorithmic structure directly into the network. This reduces the search space and guides learning toward physically meaningful solutions. As a result, these models are less dependent on large annotated datasets and can generalize more reliably across scanners and protocols. In clinical settings, this leads to more stable behaviour when acquisition conditions vary, a key requirement for deployment across institutions.

A central strength of model-based deep learning is interpretability. Unrolled architectures explicitly map each network layer to a step of an optimization algorithm, making it possible to trace how intermediate updates contribute to the final reconstruction. Physics-informed neural networks extend this idea by incorporating governing equations and boundary conditions directly into the loss function. Recent surveys emphasize that such physically constrained designs are particularly valuable in medical imaging, where transparency and consistency are essential for clinical trust and regulatory acceptance [22]. In contrast to black-box models, clinicians and developers can inspect which physical constraints were enforced, which residuals were minimized, and how these design choices influenced the output.

This interpretability is illustrated in Figure 5. The figure shows how physical laws, data fidelity terms, and neural components interact during training in a physics-informed framework. Each loss term corresponds to a well-defined modelling assumption, allowing experts to reason about failure modes and model behaviour. Such explicit structure supports error analysis and makes it easier to assess whether a model's predictions are physically plausible, an important consideration in safety-critical medical applications [23].

Model-based approaches are also more robust in challenging measurement regimes. When data are sparse, noisy, or corrupted, the embedded physics acts as a stabilizing scaffold that prevents unrealistic predictions. Purely data-driven networks, by contrast, must infer both structure and content from limited information, which often leads to hallucinated or anatomically inconsistent outputs. Although model-based methods still depend on the accuracy of their physical assumptions, these assumptions are explicit and can be inspected or refined if needed. This combination of robustness and transparency makes model-based deep learning a strong foundation for more flexible hybrid architectures, discussed in the next subsection.

C. Hybrid Mechanistic–Neural Architectures

Hybrid architectures combine physical models with neural networks by clearly separating what is known from what must be learned. A mechanistic component enforces established physiological or imaging behaviour, while a neural component learns effects that are difficult to model analytically or a priori. Clinically, this means that predictions remain grounded in known physics while still adapting to patient-specific or scanner-specific variability [25].

These approaches are especially useful when physical models are informative but incomplete. In many medical applications, governing equations describe global behaviour well, yet local tissue properties, biological variability, or acquisition specific effects deviate from idealized assumptions. Hybrid models address this gap by using the physical model as a stable backbone and learning corrections directly from data. This reduces systematic bias while preserving physically plausible behaviour.

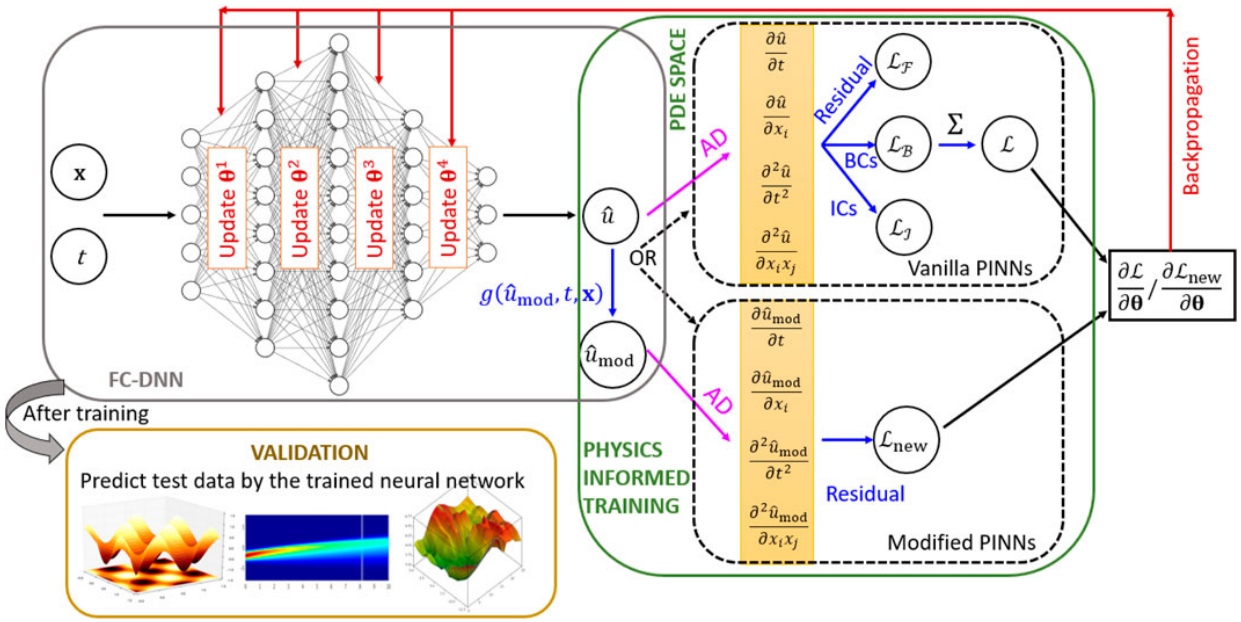


Fig. 5. Physics-informed model-based training framework illustrating the interaction between physical constraints, data fidelity, and neural components. Adapted from [24].

Designing effective hybrid systems remains challenging. If the mechanistic component is too restrictive, the neural network may be unable to correct modelling errors. If it is too weak, the model risks behaving like a black box. This balance makes explainability essential in clinical settings. Clinicians must be able to understand whether a prediction is primarily driven by trusted physical principles or by learned data-driven adjustments. These considerations naturally lead to broader questions of evaluation and clinical translation, discussed next.

D. Challenges in Comparative Evaluation and Translation

Evaluating and comparing deep learning models in medical imaging remains challenging because commonly used image-level metrics capture only a limited view of model behaviour. Measures such as PSNR and SSIM provide numerical summaries of pixel-wise similarity, but they are often insensitive to clinically relevant distortions, anatomical inconsistencies, or missing pathologies. As a result, models may achieve high scores on standard benchmarks while producing outputs that are unreliable for diagnosis.

Figure 6 illustrates this limitation through a multi-stage evaluation pipeline. Reference-based metrics quantify similarity to a ground-truth image, while non-reference metrics assess perceptual quality without explicit targets. However, both can fail to reflect performance on downstream clinical tasks such as segmentation or classification. Recent studies demonstrate that improvements in PSNR or SSIM frequently do not correlate with better task-level outcomes, highlighting a fundamental mismatch between conventional evaluation metrics and clinical relevance [26], [27].

A second challenge is robustness under changing acquisition conditions. Medical images vary substantially across scanners, institutions, and patient populations, leading to domain shift that can strongly affect model behaviour. Even model-based approaches, while generally more stable than purely data-driven methods, rely on assumptions that may not hold in all clinical workflows. In practice, this means that performance measured on curated datasets may not translate to routine hospital settings. Explainability becomes essential in this context: clinicians must understand why predictions change across conditions and whether such changes are justified.

Finally, regulatory and practical constraints further complicate clinical translation. Regulatory agencies require transparent models with clearly documented behaviour, monitoring strategies, and update mechanisms. At the same time, successful adoption depends on intuitive outputs, uncertainty estimates, and interfaces that clinicians can interpret reliably. Without these elements, even technically strong models may fail to integrate into real workflows. Together, these challenges motivate the need for unified evaluation benchmarks, multi-center validation studies, and assessment frameworks that explicitly link physical consistency to task-level clinical relevance.

VI. Challenges and Clinical Translation

Model-based deep learning (MB-DL), including unrolled networks and physics-informed architectures, promises advantages in interpretability and parameter efficiency compared to conventional deep learning methods [6]. However, even though these architectures embed domain knowledge, the clinical literature shows that their translation into real-world medical practice faces the same barriers as standard deep learning, and in some cases, MB-DL introduces unique challenges of its own. The main issues relate to dataset representative-

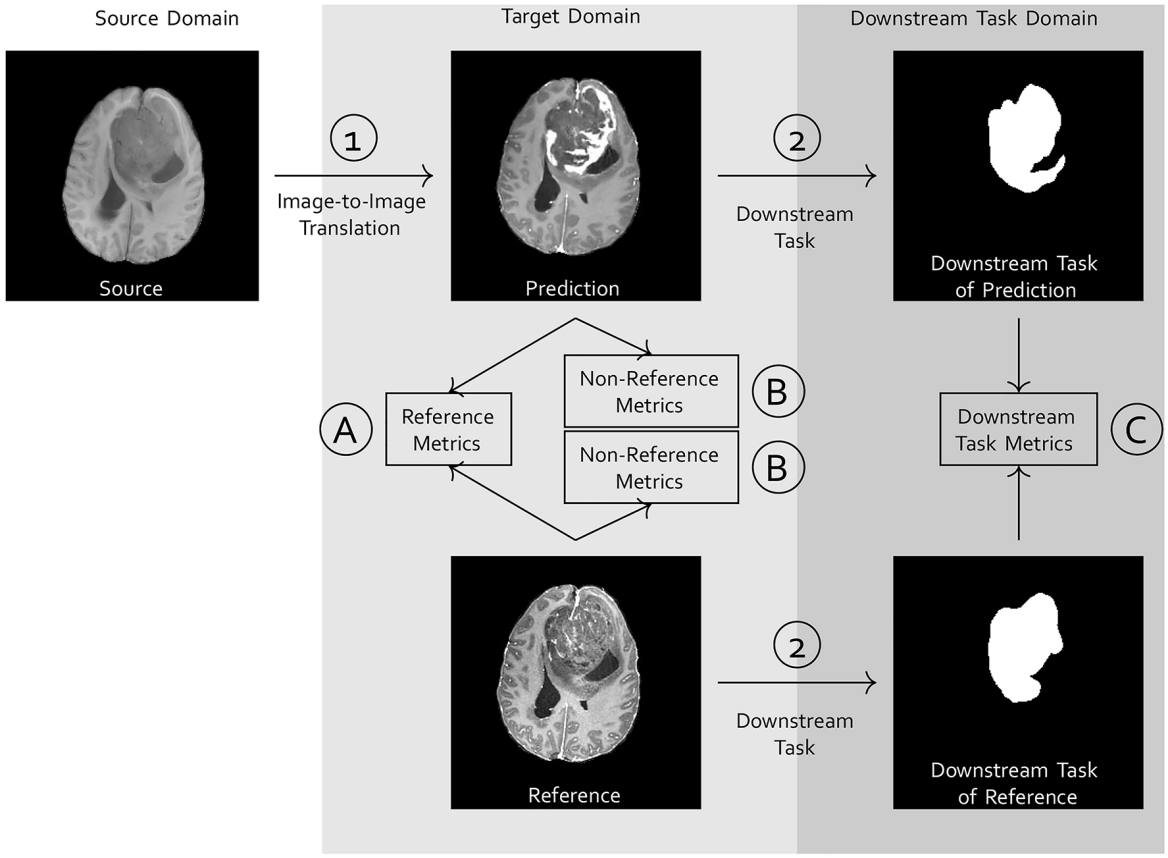


Fig. 6. Multi-stage evaluation pipeline for medical image translation, illustrating the limitations of reference-based and non-reference image quality metrics in capturing downstream clinical task performance. Adapted from [26].

ness, generalization, bias, workflow integration, and regulatory requirements. Together, these factors explain why MB-DL models that perform well in research settings often fail to reach routine clinical use.

A. Dataset Limitations and Their Impact on MB-DL

A major challenge highlighted across medical imaging studies is the limited representativeness of commonly used datasets [28], [29]. Large imaging cohorts such as the UK Biobank offer clean, homogeneous data, but they lack the variability found in clinical populations, differences in scanner hardware, acquisition protocols, disease stage, comorbidities, and artefacts.

For MB-DL, this limitation is particularly pronounced. Because these models explicitly encode physical or algorithmic priors, their performance is tightly coupled to the assumptions embedded in the model. When the training data do not capture real-world variability, the embedded model assumptions can break. For example, unrolled reconstruction networks tuned to a specific scanner or protocol may fail when the forward model in the clinic differs from the one they were designed around. Consequently, MB-DL's strengths, strong inductive biases, can become weaknesses when data shift away from those assumptions.

B. Generalization and Evaluation Practices

Several reviews show that medical imaging models are often evaluated on subsets of the same dataset used for training, rather than on external hospital cohorts [28]. This leads to overly optimistic performance estimates. Challenge competitions show the same problem: top-performing models often fail to replicate results when applied to real clinical data [30].

For MB-DL, external validation is particularly important. Because these models incorporate explicit physical models or algorithmic steps, their performance can degrade sharply when the real-world imaging pipeline diverges from the assumed one. Without rigorous multi-site testing, it is impossible to determine whether a model's embedded priors will generalize or fail silently in unseen clinical settings.

C. Bias and Fairness in Structured Architectures

Bias in medical AI, arising from demographic imbalance, scanner dependency, or protocol variation, is well documented in medical AI [29]. MB-DL does not eliminate these risks. Even though the architectures constrain learning, biased data can still lead models to pick up shortcuts (e.g., protocol-dependent signal shapes) that correlate with outcomes. Moreover, because MB-DL models explicitly encode forward models or anatomical assumptions, they may generalize even worse

than black-box models when the training distribution is skewed or narrow. Without diverse, carefully documented datasets, MB-DL cannot be deployed safely across diverse patient populations.

D. Workflow and Infrastructure Integration

Even when MB-DL models perform strongly in controlled studies, integrating them into clinical workflows remains difficult. Hospitals require compatibility with PACS systems, DICOM routing, stable inference times, and user interfaces that clinicians can trust [28], [31]. MB-DL methods, especially reconstruction or physics-based models, often depend on specific hardware configurations or tightly coupled forward models, making deployment more complex than standard classification networks.

In practice, if a system cannot ingest data from heterogeneous scanners, process it quickly enough for clinical use, or provide interpretable outputs within existing radiology systems, clinicians cannot rely on it in time-sensitive workflows.

E. Regulatory Constraints

Recent regulatory guidance, including from the FDA, emphasizes transparency, documentation of data sources, ongoing performance monitoring, and clear evidence that models generalize safely across settings [32]. MB-DL models, despite being more interpretable in theory, still require extensive documentation of their embedded physical models, assumptions, and failure modes. Ensuring safety across different imaging environments is difficult when forward models or scanner parameters vary, making regulatory approval more demanding.

F. Summary and Clinical Impact

In summary, MB-DL holds promise because it embeds physical structure and offers increased interpretability [6]. Yet clinical translation remains limited due to narrow datasets, insufficient external validation, bias, challenging workflow integration, and strict regulatory requirements [28]–[32]. These challenges matter because a model that performs well on clean research data can fail when confronted with messy, real hospital scans. If clinicians cannot rely on stable performance across scanners, protocols, and patient groups, the model cannot be safely deployed. Ultimately, the success of MB-DL in practice depends on robust multi-site validation, diverse data, clearer documentation, and deeper collaboration between clinicians, engineers, and regulatory bodies.

VII. Conclusion

Deep learning has significantly advanced medical imaging, but its safe and reliable clinical deployment requires more than high predictive accuracy. This review has shown that modelbased deep learning, through algorithm unrolling, physics-informed neural networks, and hybrid mechanistic–neural architectures, offers a principled way to embed physical structure and domain knowledge into learning-based systems. These approaches improve interpretability, data efficiency, and robustness compared to purely data-driven models, making them particularly well suited for inverse problems such as MRI reconstruction. At the same time, important challenges remain, including dataset representativeness, generalization across clinical settings, workflow integration, and regulatory approval. Addressing these issues will be essential for translating model-based deep learning from promising research prototypes into trustworthy clinical tools.

Use of AI Tools

AI-assisted tools were used during the preparation of this paper for specific supportive purposes. ChatGPT was used to assist with the structuring, refining of the search strategy, and improving the language clarity and stylistic consistency across all sections. Perplexity AI was used to support literature discovery, including identifying relevant peer-reviewed articles and publicly available figures, based on targeted search queries.

No AI system was used to generate scientific results, perform data analysis, interpret experimental outcomes, or make scientific decisions. All content, interpretations, and conclusions remain the sole responsibility of the authors.

References

[1] E. H. Houssein, A. M. Gamal, E. M. G. Younis, and E. Mohamed, "Explainable artificial intelligence for medical imaging systems using deep learning: A comprehensive review," *Springer*, vol. 28, p. 469, 2025.

[2] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, "Model-based deep learning: Key approaches and design guidelines," in *2021 IEEE Data Science and Learning Workshop (DSLW)*, 2021, pp. 1–6.

[3] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[4] D. T. Huff, A. J. Weisman, and R. Jeraj, "Interpretation and visualization techniques for deep learning models in medical imaging," *Physics in Medicine & Biology*, vol. 66, no. 4, p. 04TR01, 2021, pMCID: PMC8236074. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC8236074/#ABS1

[5] A. Q. Wang, B. K. Karaman, H. Kim, J. Rosenthal, R. Saluja, S. I. Young, and M. R. Sabuncu, "A framework for interpretability in machine learning for medical imaging," *IEEE Access*, vol. 12, pp. 53 277–53 292, 2024.

[6] V. Monga, Y. Li, and Y. C. Eldar, "Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing," *IEEE Signal Processing Magazine*, vol. 38, no. 2, pp. 18–44, 2021.

[7] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, p. 1–122, Jan. 2011. [Online]. Available: https://doi.org/10.1561/2200000016

[8] K. Gregor and Y. LeCun, "Learning fast approximations of sparse coding," in *Proceedings of the 27th International Conference on International Conference on Machine Learning*, ser. ICML'10. Madison, WI, USA: Omnipress, 2010, p. 399–406.

[9] J. Adler and O. Öktem, "Learned primal-dual reconstruction," *IEEE Transactions on Medical Imaging*, vol. 37, no. 6, pp. 1322–1332, 2018.

[10] Y. Ben Sahel, J. P. Bryan, B. Cleary, S. L. Farhi, and Y. C. Eldar, "Deep unrolled recovery in sparse biological imaging: Achieving fast, accurate results," *IEEE Signal Processing Magazine*, vol. 39, no. 2, pp. 45–57, 2022.

[11] Y. Yang, J. Sun, H. Li, and Z. Xu, "Admm-csnet: A deep learning approach for image compressive sensing," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 3, pp. 521–538, 2020.

[12] S. A. H. Hosseini, B. Yaman, S. Moeller, M. Hong, and M. Akçakaya, "Dense recurrent neural networks for inverse problems: History-cognizant unrolling of optimization algorithms," *ArXiv*, vol. abs/1912.07197, 2019. [Online]. Available: https://api.semanticscholar.org/CorpusID:209376613

[13] D. Wu, K. Kim, B. Dong, G. E. Fakhri, and Q. Li, "End-to-end lung nodule detection in computed tomography," 2018. [Online]. Available: https://arxiv.org/abs/1711.02074

[14] M. Ahmadi, D. Biswas, M. Lin, F. D. Vrionis, J. Hashemi, and Y. Tang, "Physics-informed machine learning for advancing computational medical imaging: integrating data-driven approaches with fundamental physical principles," *Artificial Intelligence Review*, vol. 58, no. 10, pp. 297–315, 2025.

[15] S. Cai, Z. Mao, J. Wang, and G. E. Karniadakis, "Physics-informed neural networks (pinns) for biomedical problems," *Journal of Computational Physics*, vol. 407, p. 109279, 2020.

[16] V. Kashtanova, M. Pop, P. Gallinari, and M. Sermesant, "Aphyn-ep: A physics-based deep learning framework to learn and forecast cardiac electrophysiology dynamics," in *Statistical Atlases and Computational Models of the Heart (STACOM)*, ser. Lecture Notes in Computer Science, 2023.

[17] S. Ahmed, J. Feng, A. Mehmood, M. U. Ali, M. Yaqub, M. A. Manan, and A. Raheem, "Pinn-dadif: Physics-informed deep adaptive diffusion network for robust and efficient mri reconstruction," *Digital Signal Processing*, vol. 160, p. 105085, 2025.

[18] S. Matta, M. Lamard, P. Zhang, A. Le Guilcher, L. Borderie, B. Cochener, and G. Quellec, "A systematic review of generalization research in medical image classification," *Computers in Biology and Medicine*, vol. 183, p. 109256, 2024.

[19] Z. Niu, S. Ouyang, S. Xie, Y.-W. Chen, and L. Lin, "A survey on domain generalization for medical image analysis," *arXiv preprint arXiv:2402.05035*, 2024.

[20] B. Guo, D. Lu, G. Szumel, R. Gui, T. Wang, N. Konz, and M. A. Mazurowski, "The impact of scanner domain shift on deep learning performance in medical imaging," *arXiv preprint arXiv:2409.04368*, 2024.

[21] O. S. Kilim, A. Olar, T. Joó *et al.*, "Physical imaging parameter variation drives domain shift," *Scientific Reports*, vol. 12, no. 21302, 2022. [Online]. Available: https://www.nature.com/articles/s41598-022-23990-4

[22] C. Banerjee, K. Nguyen, O. Salvado, T. Tran, and C. Fookes, "Pinns for medical image analysis: A survey," *arXiv preprint arXiv:2408.01026*, 2024.

[23] Z. Huang, J. Schlemper, C. Qin, J. Duan, C. Ouyang, B. Kainz, and D. Rueckert, "Model-based reconstruction with learning," *arXiv preprint arXiv:2103.14528*, 2021.

[24] T. Chatterjee, M. I. Friswell, S. Adhikari, and H. H. Khodaparast, "Matlab implementation of physics informed deep neural networks for forward and inverse structural vibration problems," *Aerospace Research Communications*, vol. 2, 2024.

[25] C. Banerjee, K. Nguyen, O. Salvado, T. Tran, and C. Fookes, "Physics-informed machine learning for medical image analysis," *ACM Computing Surveys*, vol. 56, no. 10, pp. 1–38, 2024.

[26] M. Dohmen, M. A. Klemens, I. M. Baltruschat *et al.*, "Similarity and quality metrics for MR image-to-image translation," *Scientific Reports*, vol. 15, no. 1, p. 3853, 2025. [Online]. Available: https://www.nature.com/articles/s41598-025-38530-1

[27] M. Amorós, J. Curado *et al.*, "Evaluating super-resolution models in biomedical imaging: Applications and performance in segmentation and classification," *Diagnostics*, vol. 15, no. 4, p. 1234, 2025. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC12027580/

[28] N. K. Dinsdale, L. Jenkinson, R. Jenkinson, and S. Smith, "Challenges for machine learning in clinical translation of big data imaging studies," *NeuroImage*, vol. 257, p. 119321, 2022.

[29] B. Koçak, A. Durmaz, and E. Y"ucesoy, "Bias in artificial intelligence for medical imaging: Fundamentals, detection, avoidance, mitigation, challenges, ethics and prospects," *Diagnostic and Interventional Radiology*, 2024.

[30] S. G. I. Armato, K. Drukker, and L. Hadjiiski, "Ai in medical imaging grand challenges: Translation from competition to research benefit and patient care," *The British Journal of Radiology*, vol. 96, no. 1147, p. 20230117, 2023.

[31] P. Korfiatis, T. L. Kline, H. M. Meyer, S. Khalid, T. Leiner, B. T. Loufek, D. Blezek, D. E. Vidal, R. P. Hartman, L. J. Joppa, A. D. Missert, T. A. Potretzke, J. P. Taubel, J. A. Tjelta, M. R. Callstrom, and E. E. Williamson, "Implementing artificial intelligence algorithms in the radiology workflow: Challenges and considerations," *Mayo Clinic Proceedings: Digital Health*, vol. 3, no. 1, p. 100188, 2025.

[32] U.S. Food and Drug Administration, "Artificial intelligence-enabled device software functions: Lifecycle management and marketing submission recommendations," Jan. 2025, draft Guidance.